Sanjeevani (SahayAI)

Comprehensive System Architecture, Technical Deep-Dive & Evaluation Defense Dossier

Sanjeevani (SahayAI): Comprehensive Project Guide & Defense Dossier

> **Emergency Care Decision Support & Live Bed Coordination Platform**
> *Developed for Anvathon 2026, NIE Mysuru*

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1. Executive Summary

In acute medical crises—such as severe trauma, acute ischemic stroke, pediatric seizures, extensive burns, or venomous snakebites—survival depends on the **"Golden Hour"**. Today, panicked patients and families rely on conventional map applications (e.g., Google Maps) that optimize strictly for **geographic proximity**.

This frequently leads to catastrophic delays: a family drives 10 minutes to the nearest small clinic or dispensary only to discover it lacks a pediatric ICU, cardiac catheterization lab, burns unit, or antivenom stock. They are turned away and forced to navigate to a tertiary center, losing irreversible time.

Sanjeevani solves this fundamental failure mode:
1. **Capability-First Triage**: Matches patient symptoms to specialized hospital clinical facilities (Cath Lab, PICU, Stroke unit, Antivenom, Trauma resuscitation), not distance alone.
2. **Nearest Stabilization Fallback**: If specialized care is unavailable within reach, immediately routes the patient to the nearest 24×7 emergency resuscitation center for initial stabilization before inter-hospital transfer.
3. **Live Departmental Bed Visibility**: Tracks live bed availability across critical care units (Emergency Casualty, ICU, CCU, PICU, NICU, Burns, General) with an authenticated hospital staff admin portal (`/admin`).
4. **Offline Multilingual Natural Interaction**: Supports English, Hindi (Devanagari & Romanized), and Kannada (Script & Romanized) via offline local speech-to-text (`faster-whisper`) and deterministic rule engines.
5. **Immediate Care Coordination**: Generates structured clinical Pre-Alert text for casualty staff handoff,

1-click direct emergency phone calling (`tel:`), coordinate-pinned Google Maps navigation, community helper alerts, and a local ambulance network.

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2. System Architecture

```
flowchart TD
    subgraph ClientLayer["Frontend & Client Layer (Next.js 16 + React 19)"]
        UI["Responsive Web UI / PWA\n(Sanjeevani App)"]
        VoiceRec["Microphone Audio Recorder\n(Web Audio API / MediaRecorder)"]
        AdminUI["Staff Live Bed Portal\n(/admin)"]
        Geo["Browser Geolocation / Selected Mysuru Area"]
    end

    subgraph APILayer["API & Security Gateway (FastAPI + Pydantic v2)"]
        CORS["CORS & Rate Limiter\n(60 req/min per IP)"]
        TriageRoute["POST /api/triage"]
        VoiceRoute["POST /api/voice/transcribe"]
        BedsRoute["GET/PUT /api/admin/hospitals/:id/beds"]
        NearbyRoute["POST /api/nearby/helper & /request"]
    end

    subgraph ProcessingLayer["Engine & Intelligence Core"]
        WhisperEngine["Local Faster-Whisper\n(CTranslate2 CPU INT8 Fallback)"]
        Classifier["Deterministic Trilingual Classifier\n(English / Hindi / Kannada Lexicon)"]
        Guard["Unrecognized Phrase Guard\n(Gating Non-Emergency Phrases)"]
        RankingEngine["Capability & Bed-Aware Scorer\n(Haversine Distance + Clinical Taxonomy)"]
        PrealertGen["Clinical Pre-Alert Generator"]
    end

    subgraph DataLayer["Storage & Data Pipeline"]
        HospitalDB["Mysuru Hospitals & Bed State\n(data/hospitals.json)"]
        TaxonomyDB["Clinical Capability Taxonomy\n(data/taxonomy.json)"]
        AmbulanceDB["Ambulance Directory\n(data/mysuru_ambulance_providers.json)"]
        SQLiteDB["Nearby Helper Alerts DB\n(.data/nearby.sqlite3)"]
    end

    VoiceRec -->|Raw Audio Stream| VoiceRoute
    VoiceRoute --> WhisperEngine
    WhisperEngine -->|Transcribed Text| UI

    UI -->|Triage Request + GPS| CORS
    AdminUI -->|Live Bed Updates| CORS
    CORS --> TriageRoute
    CORS --> BedsRoute
    CORS --> NearbyRoute

    TriageRoute --> Classifier
    Classifier --> Guard
    Guard -->|If Unrecognized| UI
    Guard -->|If Medical Emergency| RankingEngine

    RankingEngine --> HospitalDB
    RankingEngine --> TaxonomyDB
    RankingEngine --> PrealertGen
    PrealertGen -->|Structured Match + PreAlert| UI
```

```
    BedsRoute --> HospitalDB
    NearbyRoute --> SQLiteDB
```

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3. End-to-End Workflow

```
sequenceDiagram
    autonumber
    actor Patient as Patient / Bystander
    participant Web as Sanjeevani UI (Next.js)
    participant API as FastAPI Backend (:8000)
    participant AI as Whisper & Classifier
    participant Rank as Capability Ranking Engine
    actor Staff as Hospital Staff / ER Nurse

    %% Voice or Text Input
    alt Spoken Input
        Patient->>Web: Clicks Mic & Speaks ("ನನ್ನ ಸಹೋದರನಿಗೆ ಹಾವು ಕಚ್ಚಿದೆ")
        Web->>API: POST /api/voice/transcribe (raw audio)
        API->>AI: Faster-Whisper (Kannada acoustic model)
        AI-->>API: Transcribed Text & Detected Language
        API-->>Web: "ನನ್ನ ಸಹೋದರನಿಗೆ ಹಾವು ಕಚ್ಚಿತು"
    else Typed Input
        Patient->>Web: Types emergency description
    end

    %% Triage Execution
    Web->>API: POST /api/triage {text, lat, lng, language}
    API->>AI: Classify emergency type, severity, red flags

    alt Non-Emergency Input (e.g. "hello", "i am going")
        AI-->>API: type = general_unclear, confidence = 0.0, red_flags = []
        API-->>Web: recommendation = null, warnings = ["Unrecognized phrase"]
        Web-->>Patient: Displays "Could not understand description" guide + 108/112 buttons
    else Medical Emergency (e.g. "Snake bite")
        AI-->>API: type = snake_bite, severity = critical
        API->>Rank: Rank hospitals against taxonomy requirements

        alt All Specialized Capabilities Met Nearby
            Rank-->>API: Returns Ranked Hospital (e.g. St. Joseph's / JSS Hospital)
        else Missing Specialized Capabilities
            Rank-->>API: Fallback to nearest 24x7 Emergency for Stabilization
        end

        API-->>Web: Hospital match, department beds, Google Maps link, Pre-Alert text
        Web-->>Patient: Displays Recommended Hospital + Bed Availabilities + Actions
        Patient->>Web: Clicks "Copy Pre-Alert" & "Call Hospital"
    end

    %% Live Bed Coordination
    Staff->>Web: Accesses /admin & Updates Bed Counts
    Web->>API: PUT /api/admin/hospitals/hosp_002/beds
    API->>API: Validates capacity (available <= total) & persists state
    API-->>Web: Updated live bed counts reflected instantly across users
```

4. Technical Stack & Deep Dive

A. Frontend Architecture

- **Framework**: Next.js 16.4 (App Router) with React 19.
- **Language & Typing**: TypeScript with zero `any` tolerance; strict schema validation via **Zod** contracts (`src/features/patient/lib/contracts.ts`).
- **Styling**: Tailored responsive CSS tokens with accessibility contrast ratios, fluid layouts for mobile browsers, and reduced-motion media query respect.
- **State Management**: Reactive local hooks with optimistic UI transitions and clipboard fallback polyfill for WebKit/Android WebView edge cases.
- **PWA & Offline Capability**: Web App Manifest (`manifest.webmanifest`), service-worker ready, mobile home-screen installable.

B. Backend Architecture

- **Framework**: FastAPI (Python 3.12) with Uvicorn ASGI server.
- **Serialization & Validation**: Pydantic v2 compiled core for sub-millisecond serialization and strict schema boundaries (`extra="forbid"` on requests).
- **Concurrency**: Pure non-blocking `async` / `await` architecture.
- **Rate Limiting & Safety**: Token-bucket client IP limiter (60 requests/minute) preventing malicious exhaustion during emergency loads.
- **CORS**: Configured for local dev, IP network testing (`0.0.0.0`), and production domains with preflight options.

C. Offline Speech-to-Text (`faster-whisper`)

- **Engine**: CTranslate2 engine leveraging OpenAI Whisper models.
- **Resilience**: Dynamic CPU INT8 quantization fallback if CUDA libraries (`cublas64_12.dll`) are absent on host machines.
- **Audio Handling**: Supports multi-format incoming raw audio (`audio/webm` , `audio/wav` , `audio/ogg` , `audio/mp4`) with memory bounds checking (max 10MB, max 60s) to prevent memory-spike DoS.

D. Data Pipeline & Persistence

- **Hospital Dataset**: 23 verified/public hospital records across Mysuru urban and rural districts (`data/hospitals.json`).
- **Medical Capability Taxonomy**: Clinical taxonomy mapping acute medical conditions to required vs. preferred infrastructure (`data/taxonomy.json`).
- **Department Bed Breakdown**: Synthesized initial bed inventories across 7 departments per hospital, dynamically updated in-memory with atomic file write-through.

- **Community Alerts**: Local SQLite (`.data/nearby.sqlite3`) for storing short-lived, privacy-preserving nearby helper registrations and push notifications.

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5. Architectural Decision Matrix: "Why We Chose X Over Y"

Component / Layer	Chosen Technology	Alternatives Considered	Why Chosen Over Alternatives
Backend Framework	**FastAPI (Python 3.12)**	Django, Flask, Express.js	FastAPI runs native `asyncio`, utilizes Rust-compiled Pydantic v2 for sub-millisecond serialization, auto-generates OpenAPI docs, and natively integrates with Python ML/NLP tools. Django is overly monolithic; Flask lacks built-in async validation; Express lacks native Python AI library bindings.
Frontend Framework	**Next.js 16 (App Router)**	Vite + React SPA, Create React App	Next.js provides server prerendering for near-zero first-contentful-paint (critical on spotty 3G/4G cellular connections in transit), built-in asset optimization, and unified TypeScript contract sharing.
Voice Transcription	**Faster-Whisper (Local CPU INT8)**	Google Cloud STT, OpenAI Cloud Whisper API	In an emergency, relying on third-party cloud APIs introduces network round-trip latency (1–3s), network failure vulnerability, and exposes raw patient voice audio. Local `faster-whisper` runs offline in ~400ms on CPU with 100% data privacy.
Triage & Classification	**Hybrid Rule Lexicon + LLM Fallback**	Pure LLM (e.g. Cloud GPT-4), Simple Regex	A pure LLM is nondeterministic, can hallucinate hospital capabilities, and adds 2–4s latency. Regex is too brittle for Romanized Kannada/Hindi. Our hybrid lexicon matches in <2ms with 100% determinism, reserving LLMs as an optional background assistant.
Hospital Ranking	**Capability-First Haversine Scoring**	Google Places API, Distance-only Maps API	Google Places API ranks by commercial relevance and proximity; it cannot distinguish whether an ER has a Pediatric ICU, Snake Antivenom, or Burns ICU. Our custom engine filters by medical capability first and optimizes distance second.
Live Bed State	**In-Memory Cache with Atomic JSON Sync**	PostgreSQL, MongoDB, Redis	Eliminates complex external daemon dependencies during hackathon presentation while delivering sub-millisecond read/write latency. Changes persist reliably across server restarts without DB connection pool exhaustion.

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6. Detailed Feature Breakdown

1. Capability & Bed-Aware Triage Scoring

Hospitals are evaluated dynamically on every request:
$$\text{Score} = (\text{Capability Match Score}) - (\text{Distance Penalty}) + (\text{Bed Headroom Bonus})$$
* **Required Capabilities**: Must be 100% satisfied. If missing, hospital is excluded from primary recommendations.
* **Preferred Capabilities**: Adds bonus points for specialized facilities (e.g., PICU for pediatric patients).
* **Distance Decay**: Log-damped penalty proportional to calculated driving travel time (assuming average 25–40 km/h Mysuru traffic).
* **Live Beds Headroom**: Hospitals with confirmed available beds in the required department receive prioritization over overcrowded hospitals.

2. Stabilization Fallback Mechanism

- If an incident requires rare capabilities (e.g., Hyperbaric Chamber or Specialized Plastic Surgery Burns Unit) and no nearby hospital meets all requirements:
- The system automatically activates `no_full_match = True`.
- Gathers all nearby hospitals possessing **24×7 Emergency Resuscitation (`emergency_24x7`)**.
- Ranks candidates strictly by shortest driving time.
- Issues an explicit clinical warning:
  "No hospital listing all specialized capabilities needed for this emergency was found nearby. Showing nearest hospitals with 24×7 emergency care for initial patient stabilization. Call 108."

3. Graceful Non-Emergency Input Gating

- Random inputs (e.g., "*hello*", "*i am going home*", "*testing 123*") are prevented from returning false matches.
- When `type == general_unclear` and `confidence == 0.0` with no clinical red flags:
- Recommendation and alternatives are set to `null` / `[]`.
- Pre-alert text is cleared.
- Renders a distinct "*Could not understand description*" guidance card prompting the user to describe medical symptoms or tap direct 108/112 hotline buttons.

4. Hospital Staff Live Bed Management Portal (`/admin`)

- Accessible directly at `/admin`.
- Searchable and filterable by hospital name, locality, or department.
- Real-time counters showing total capacity vs. live available beds across Emergency Casualty, ICU, CCU, PICU, NICU, Burns, and General wards.

- Validates bed updates on client and backend (`available_beds <= total_beds`).
- Updates immediately propagate to the triage search engine.

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7. Presentation Slide Deck Structure (PPT Blueprint)

Use this 8-slide structure for your presentation:

Slide 1: Title & The Golden Hour Dilemma

- **Title**: Sanjeevani (Right Hospital. First Time.)
- **Subtitle**: Capability-First Emergency Triage & Real-Time Hospital Bed Coordination
- **The Problem**: In acute emergencies, Google Maps routes patients to the *nearest* facility, not the *equipped* facility. Arriving at a clinic without a Cath Lab or Antivenom costs lives during the Golden Hour.

Slide 2: The Solution: Sanjeevani

- **Capability-First Routing**: Matching patient condition to clinical infrastructure.
- **Trilingual Accessibility**: Kannada, Hindi, and English support for voice and text.
- **Live Bed Transparency**: Departmental bed counts visible to patients and updatable by staff.
- **Pre-Alert Coordination**: Notifying casualty staff before the ambulance arrives.

Slide 3: System Architecture & Data Flow

- *Include the Mermaid Architecture Diagram from Section 2.*
- Emphasize the clean separation between Next.js frontend, FastAPI async backend, and offline Whisper STT.

Slide 4: Real-World Scenario Walkthrough

- **Scenario A: Viper Snakebite in Rural Mysuru**:
- Input: Spoken Kannada "ಹಾವು ಕಚ್ಚಿದೆ"
- Engine detects `snake_bite` emergency (`critical`), queries antivenom + ICU capability, routes to St. Joseph's / JSS Hospital.
- Formats pre-alert text ready for casualty desk.
- **Scenario B: Casual Input / Stabilization Fallback**:
- Input "*hello*" -> Clean guidance card without fake recommendations.
- Rare condition -> Fallback to nearest 24×7 hospital for patient stabilization.

Slide 5: Hospital Staff Admin Portal (`/admin`)

- *Show screenshot of* `/admin` *portal.*
- Department-level granularity (ICU, PICU, CCU, Burns, Casualty).
- Instant sync between hospital staff updates and patient triage results.

Slide 6: Technical Innovations & Resilience

- **Zero-Cloud Offline Voice**: Faster-Whisper running on local CPU with INT8 quantization.
- **Sub-5ms Triage Latency**: Deterministic trilingual clinical lexicon.
- **Safety & DoS Protection**: Rate limiting, strict Pydantic schemas, and audio memory guards.

Slide 7: Live Demo Highlights

- Walk through 3 quick actions:
- Voice transcription in local language.
- Bed availability breakdown card.
- Live staff bed count update in `/admin` updating triage recommendations instantly.

Slide 8: Future Roadmap & Impact

- Integration with Karnataka 108 CAD (Computer Aided Dispatch).
- Direct HL7 / FHIR EHR data pipelines.
- IoT ambulance telemetry (vital signs streaming in transit).

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8. High-Probability Q&A Defense Guide

Q1: "Why don't you just use Google Maps or Google Places API?"

> **Answer**:
> "Google Maps is a commercial discovery platform optimized for distance, ratings, and business hours. It has zero awareness of clinical medical capabilities. For example, Google Maps cannot tell a bystander whether a hospital has a pediatric intensive care unit (PICU), snake antivenom stock, or a specialized burns ICU with an available bed. Sanjeevani uses clinical taxonomy and departmental telemetry to ensure patients go to the right hospital the first time."

Q2: "What if a user types something completely unrelated like 'hello' or 'where is the bus station'?"

> **Answer**:
> "Earlier prototypes of emergency triage tools often returned a default hospital for any text. In Sanjeevani, we built an explicit Unrecognized Phrase Guard. When text contains zero medical keywords or red flags, confidence is 0.0, and the system explicitly suppresses hospital recommendations and pre-alerts. It presents a dedicated explanatory card instructing the user to describe clinical symptoms or call 108/112."

Q3: "What happens if a patient has an emergency where no hospital nearby has the specialized equipment?"

> **Answer**:
> "We implemented an automatic Stabilization Fallback protocol. When no hospital meets 100% of the specialized criteria, rather than failing or sending the patient far away, the algorithm immediately identifies the nearest facility with 24×7 emergency resuscitation capabilities (`emergency_24x7`). It routes the patient there for immediate vital sign stabilization while displaying a clear clinical warning advising transfer coordination."

Q4: "Why did you use local Whisper instead of Google Cloud Speech-to-Text or OpenAI API?"

> **Answer**:
> "In high-stakes emergencies, two things are non-negotiable: latency and data privacy. External cloud APIs add 1 to 3 seconds of network round-trip overhead and fail entirely during cellular outages. By deploying `faster-whisper` locally with CTranslate2 and CPU INT8 quantization, audio is transcribed in under 400ms directly on the machine without patient voice data ever leaving the secure local perimeter."

Q5: "How do you prevent hospital bed numbers from becoming stale?"

> **Answer**:
> "We provide an ultra-lightweight Hospital Staff Portal at `/admin` designed for casualty desk charge nurses. Bed counts are updated with a single tap on mobile or desktop via authenticated REST endpoints. In our next phase, this API integrates directly with hospital HIS/EHR databases via standard FHIR bed-availability protocols."

Q6: "How did you ensure system stability under hackathon demo conditions?"

> **Answer**:
> "The entire codebase is verified by an automated test suite: 12 Node.js contract tests, strict TypeScript compilation, ESLint, Prettier formatting, and 16 Python backend unit tests. In addition, our FastAPI backend features client IP rate limiting (60 requests/minute) and strict payload validation to prevent crashes."